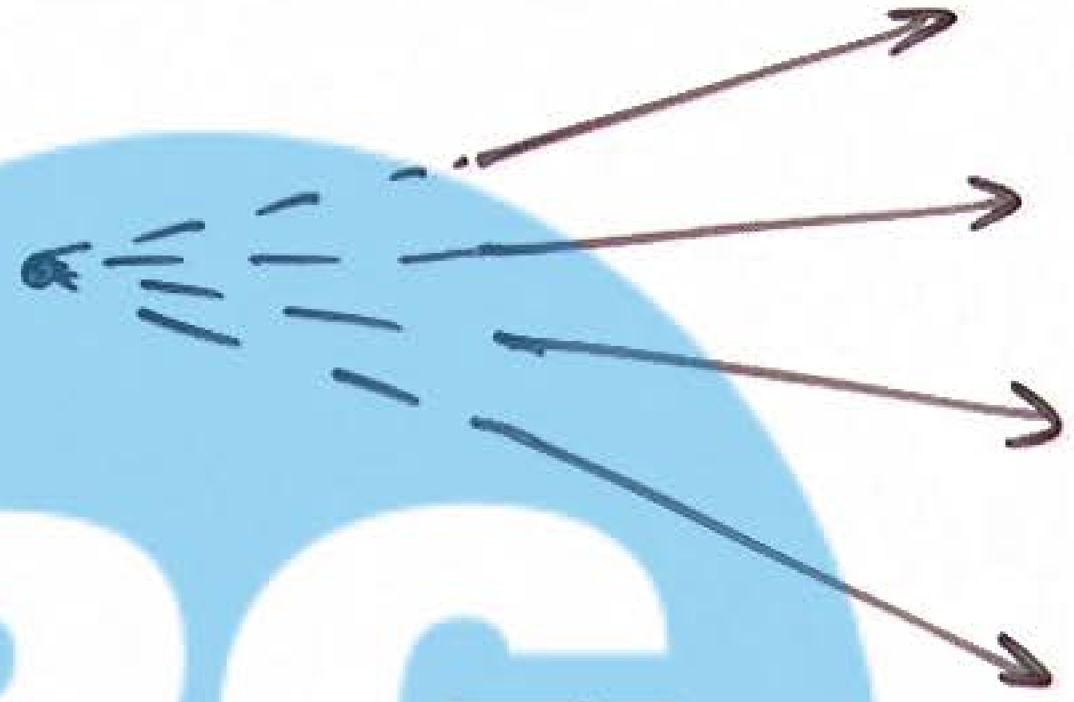
REFLECTION OF LIGHT

① Ray Optics :

Parallel beam of light

Ray → V. thin beam of light
→ It is a theoretical Concept.



Diverging Beam

Beam. → Bundle of several light Rays

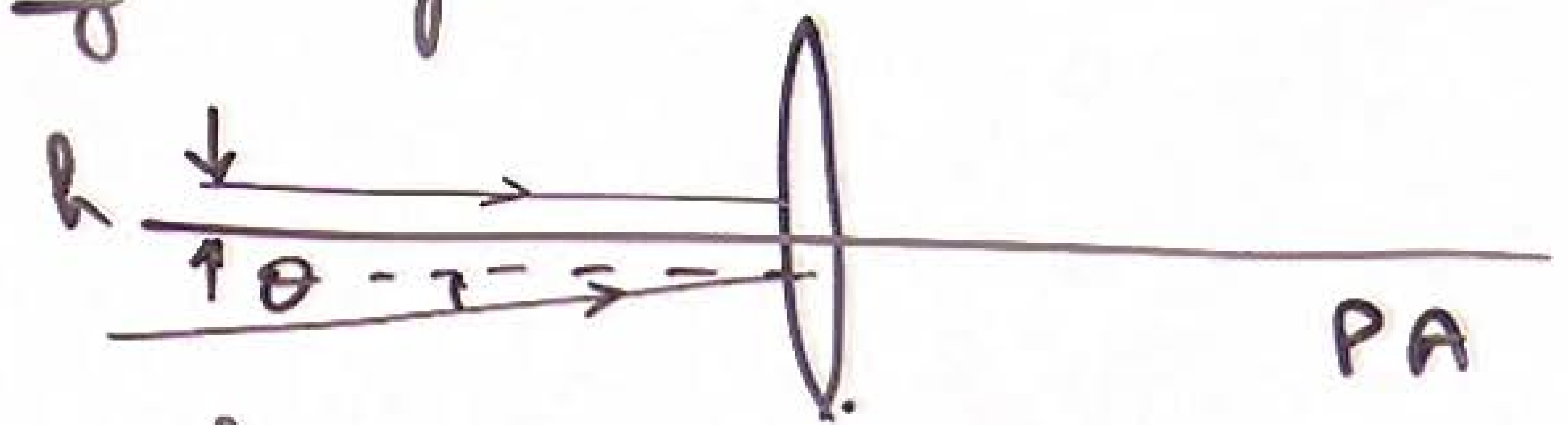
Converging Beam.

① Ray Optics :

Ray → V. thin beam of light

→ It is a theoretical Concept.

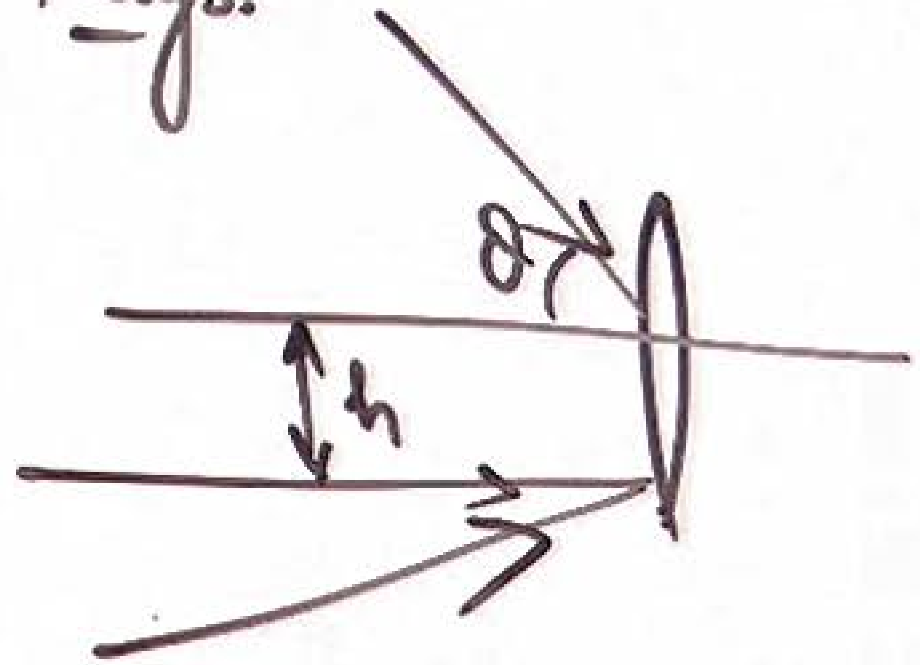
→ Paraxial light Rays → Rays very close and almost parallel to PA.



h, θ is v. small

→ Marginal light Rays.

h, θ → can be large.

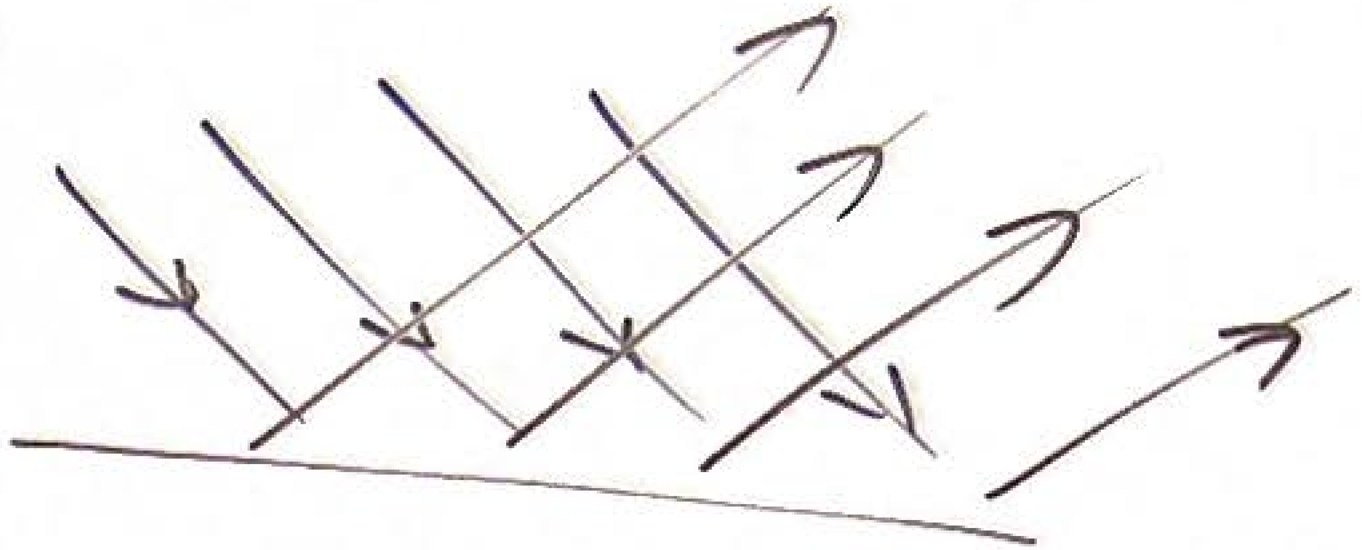


Beam. → Bundle of several light Rays

REFLECTION OF LIGHT

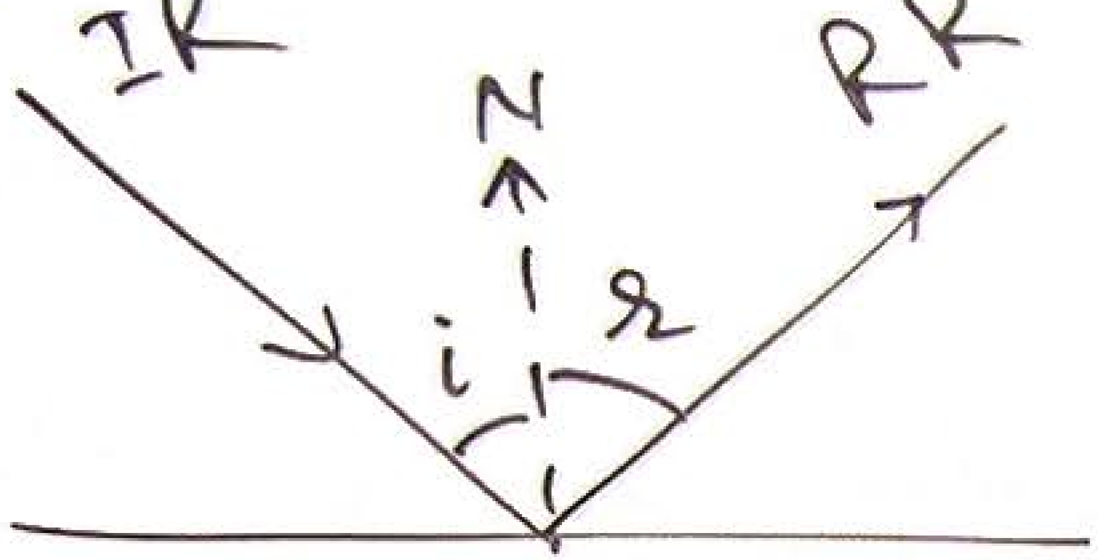
(2) Reflection of Light :

→ Specular Reflection
(Regular)



Laws of Reflection

(1)

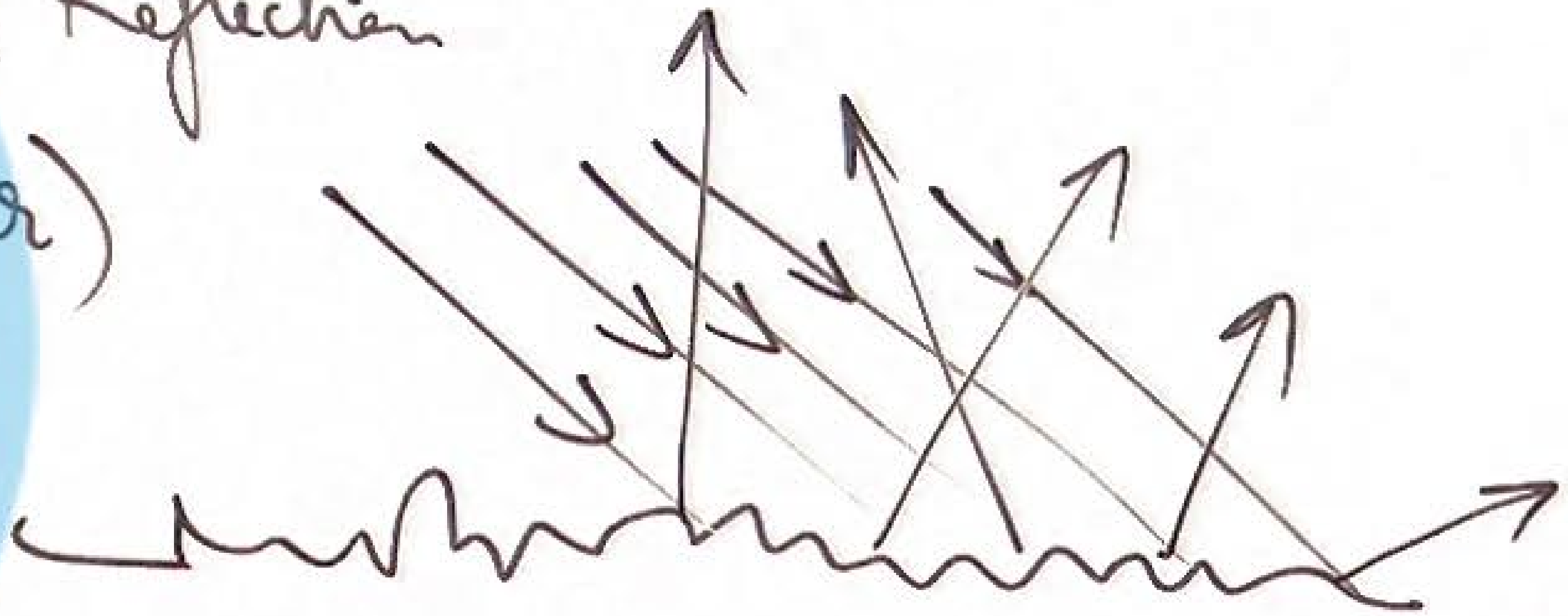


$\angle i = \angle r$.

(2) IR, RR, N lie in Same plane.

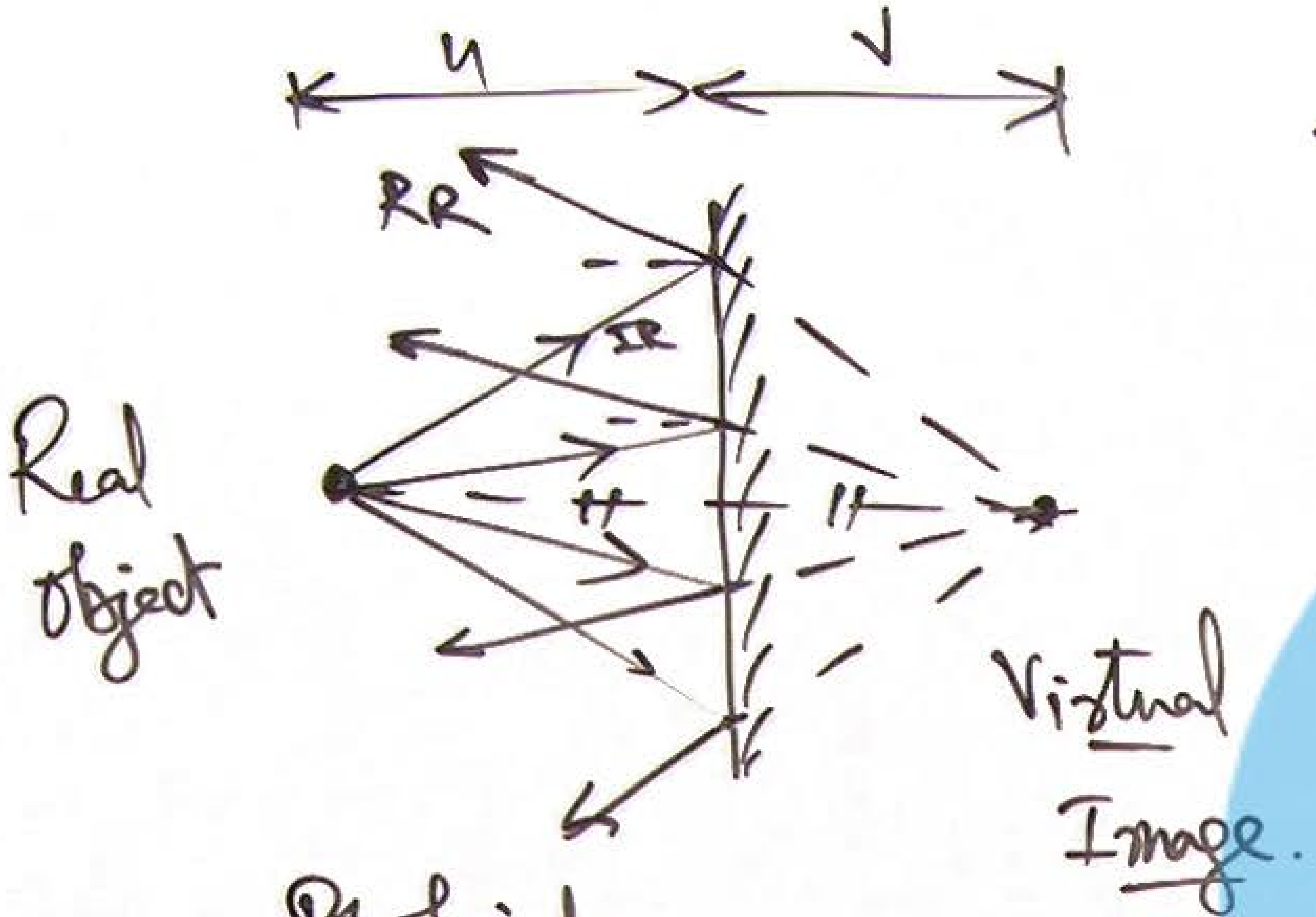
$$\underbrace{\hat{IR} \times \hat{n} = \hat{RR} \times \hat{n}}$$

Diffused Reflection
(Irregular)



REFLECTION OF LIGHT

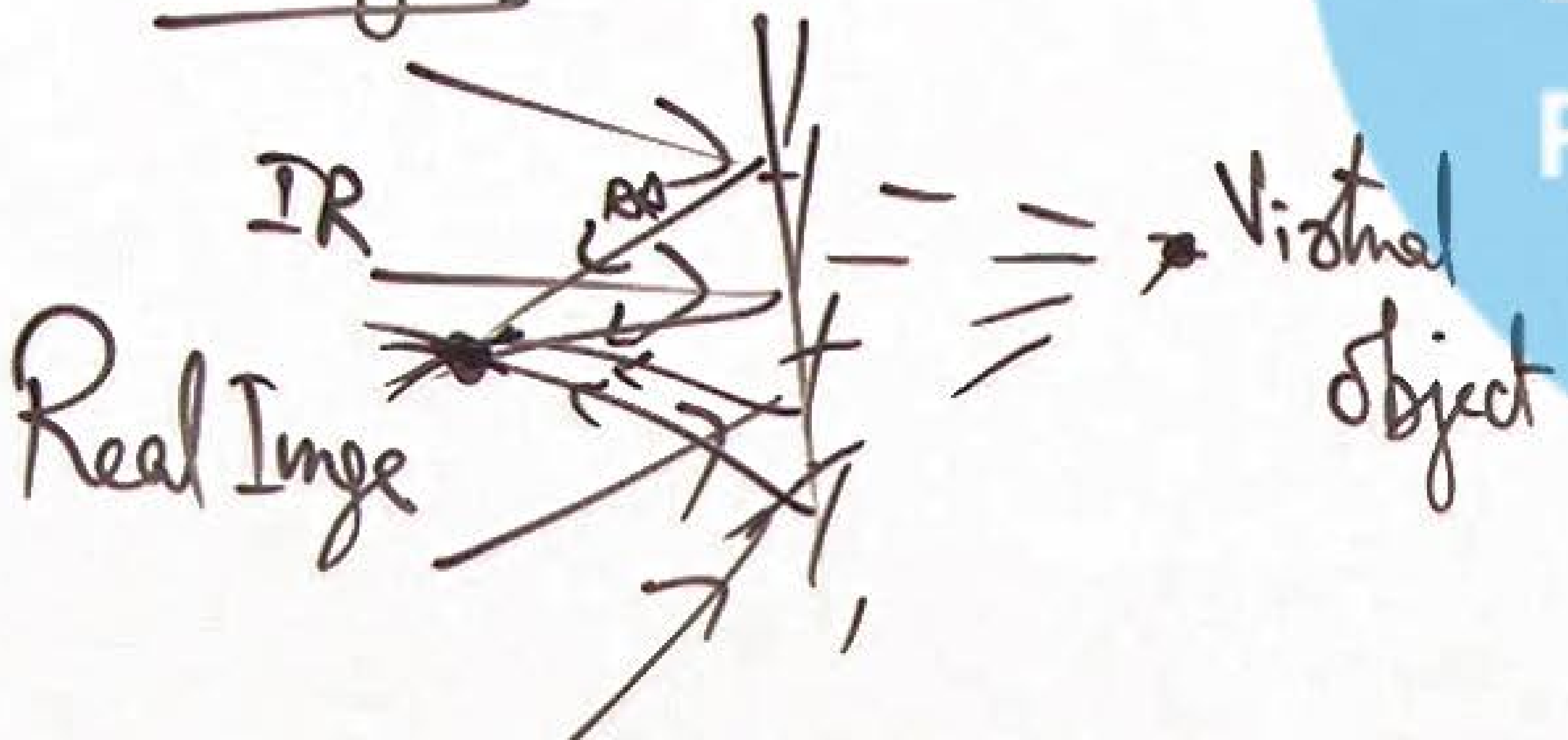
(3) Image Formation by Plain Mirror:



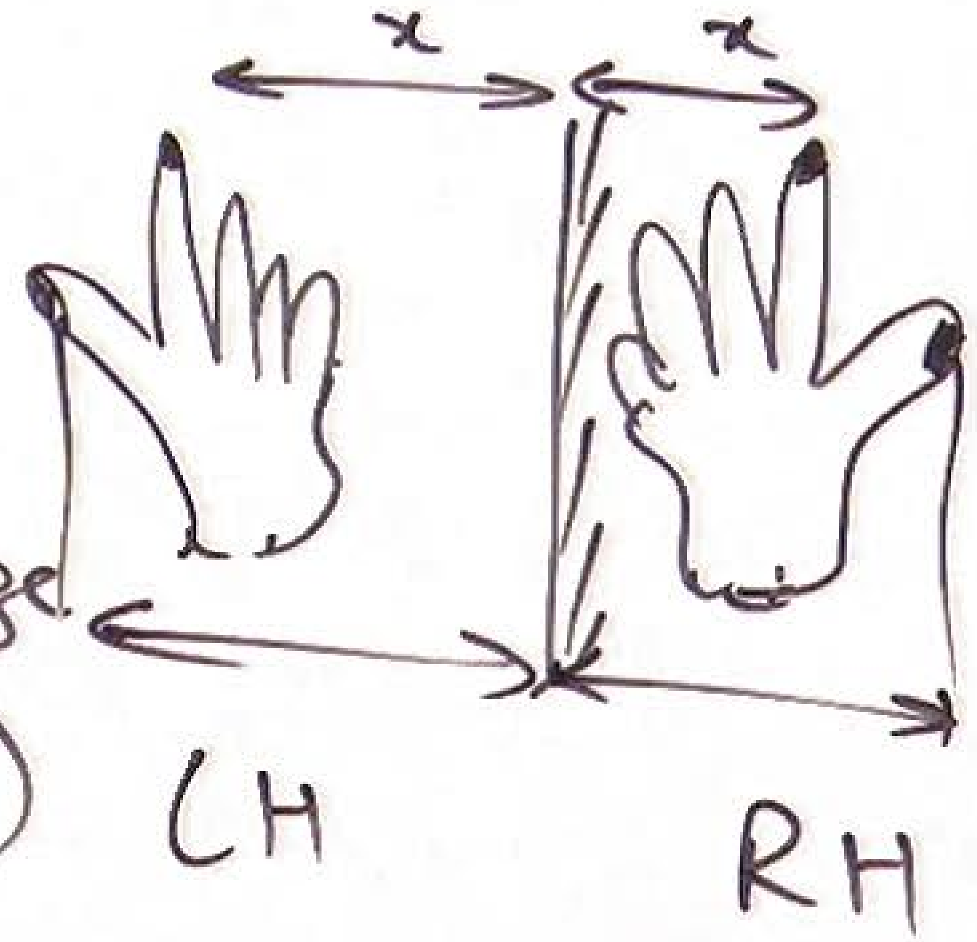
→ object dist = Image dist

→ nature of object in (R/V) opp to nature of Image (V/R)

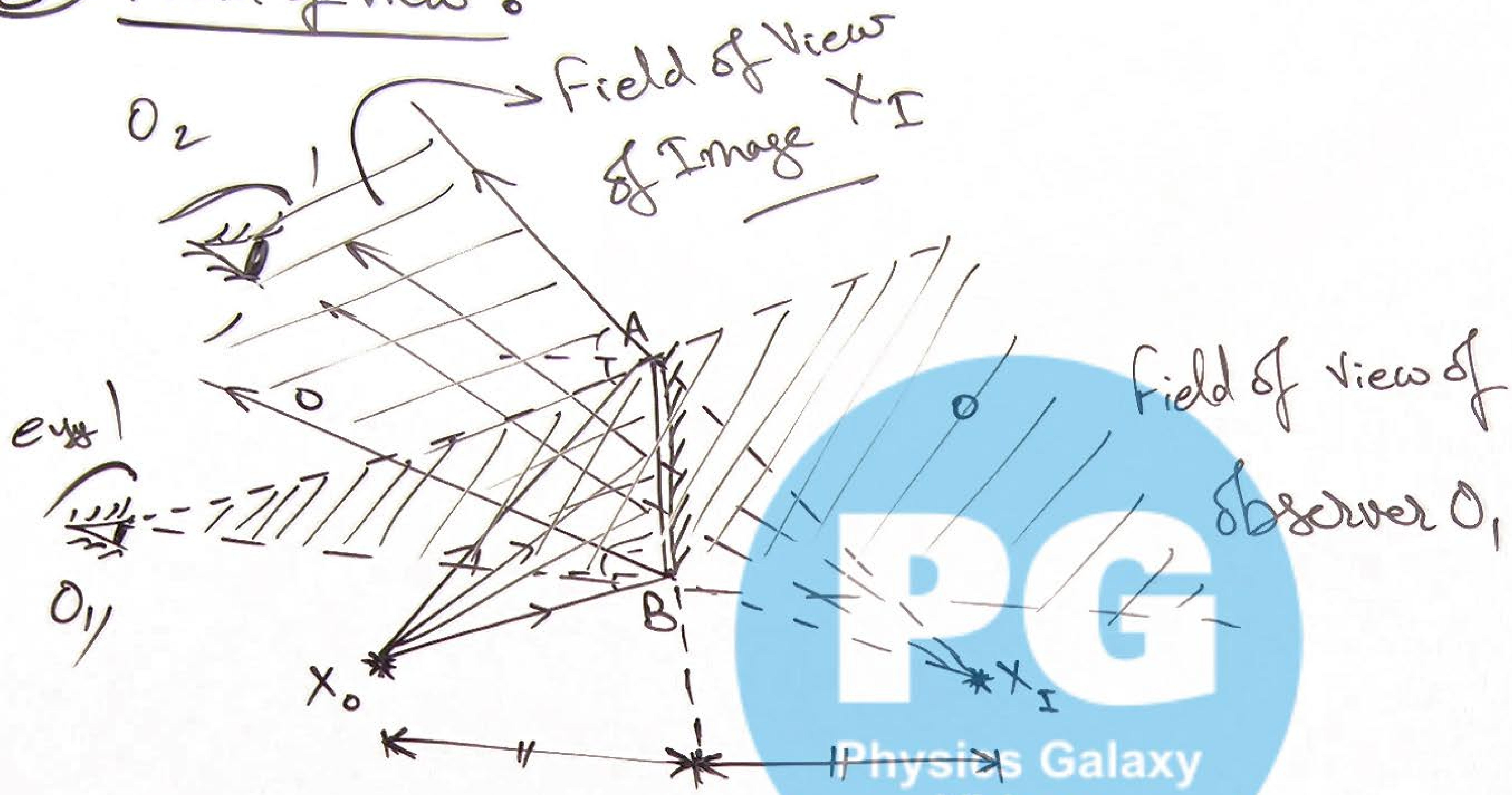
Pt object



(4) Lateral Inversion by Plane Mirror

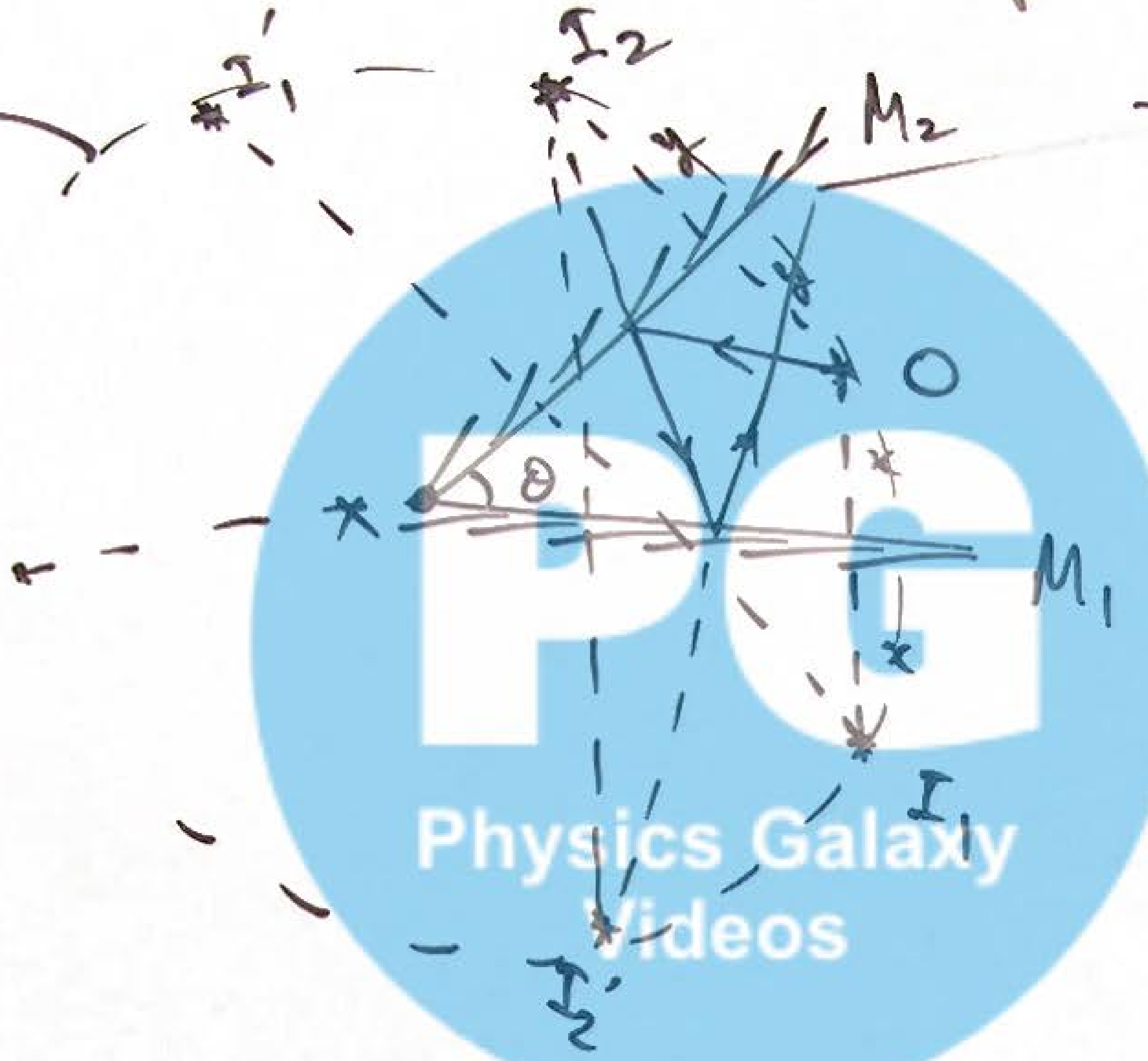


(5) Field of View :



6 Image formation by Inclined Mirrors:

All multiple images lie on a circle.



if $\theta = 60^\circ \Rightarrow N = 5$

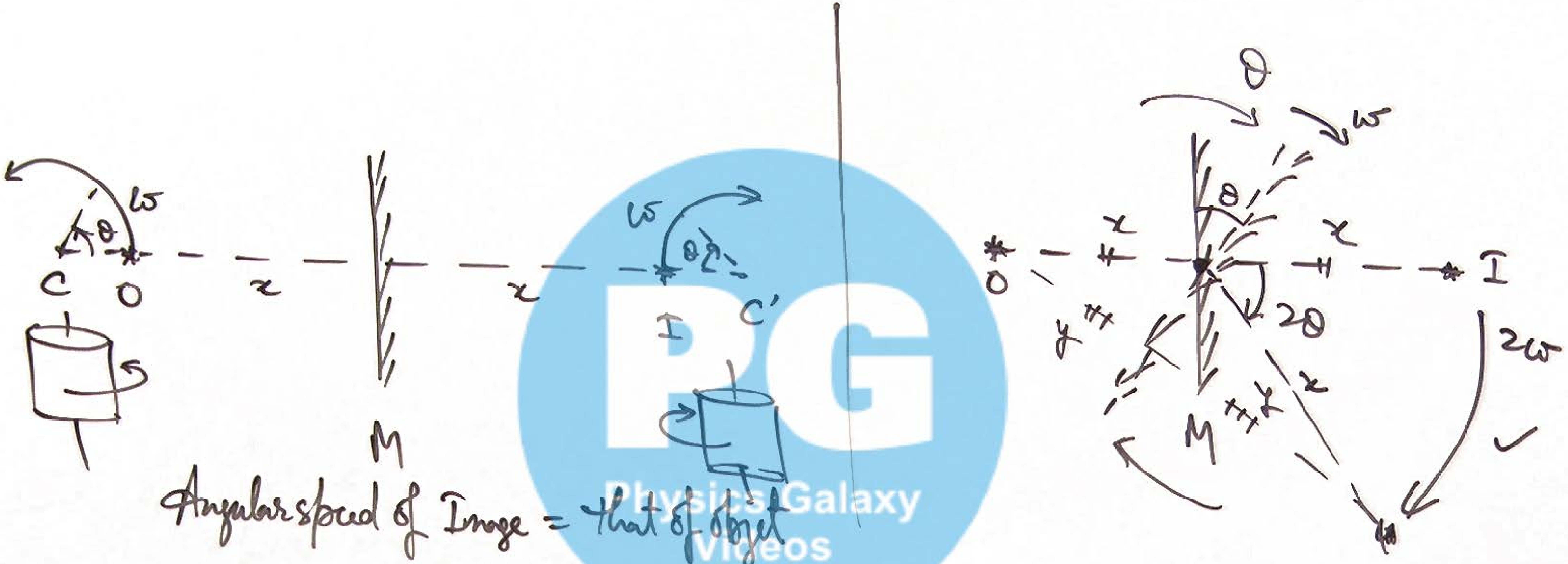
if $\theta = 59^\circ$ or $61^\circ \Rightarrow N = 7$.

if $\frac{180}{\theta} \in I$

then no of Image

$$\boxed{N = \frac{360}{\theta} - 1}$$

(7) Effect of Rotation of Object & Mirror:

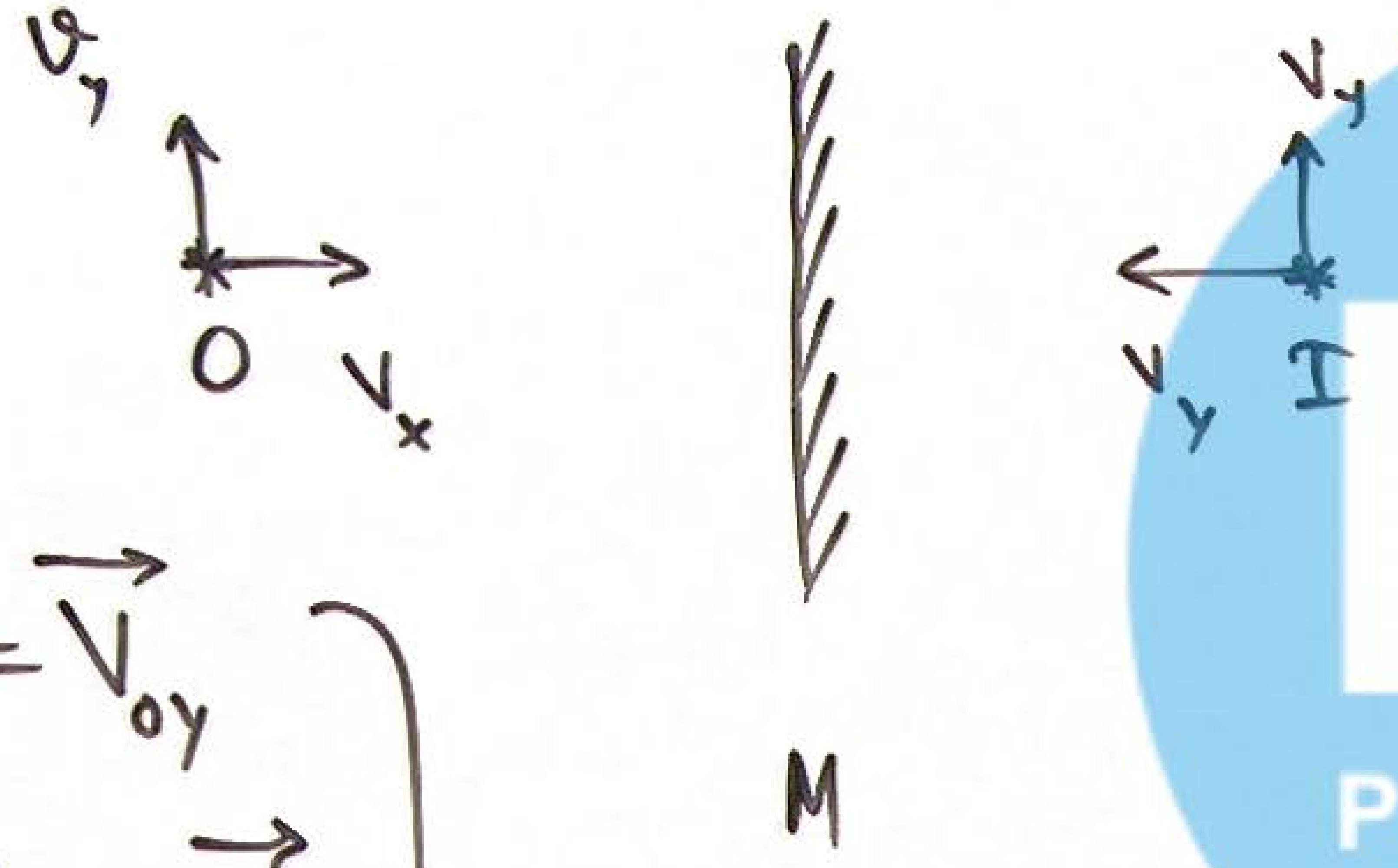


Angular speed of Image = that of object

dir of rot^n/rev is opp to that of object

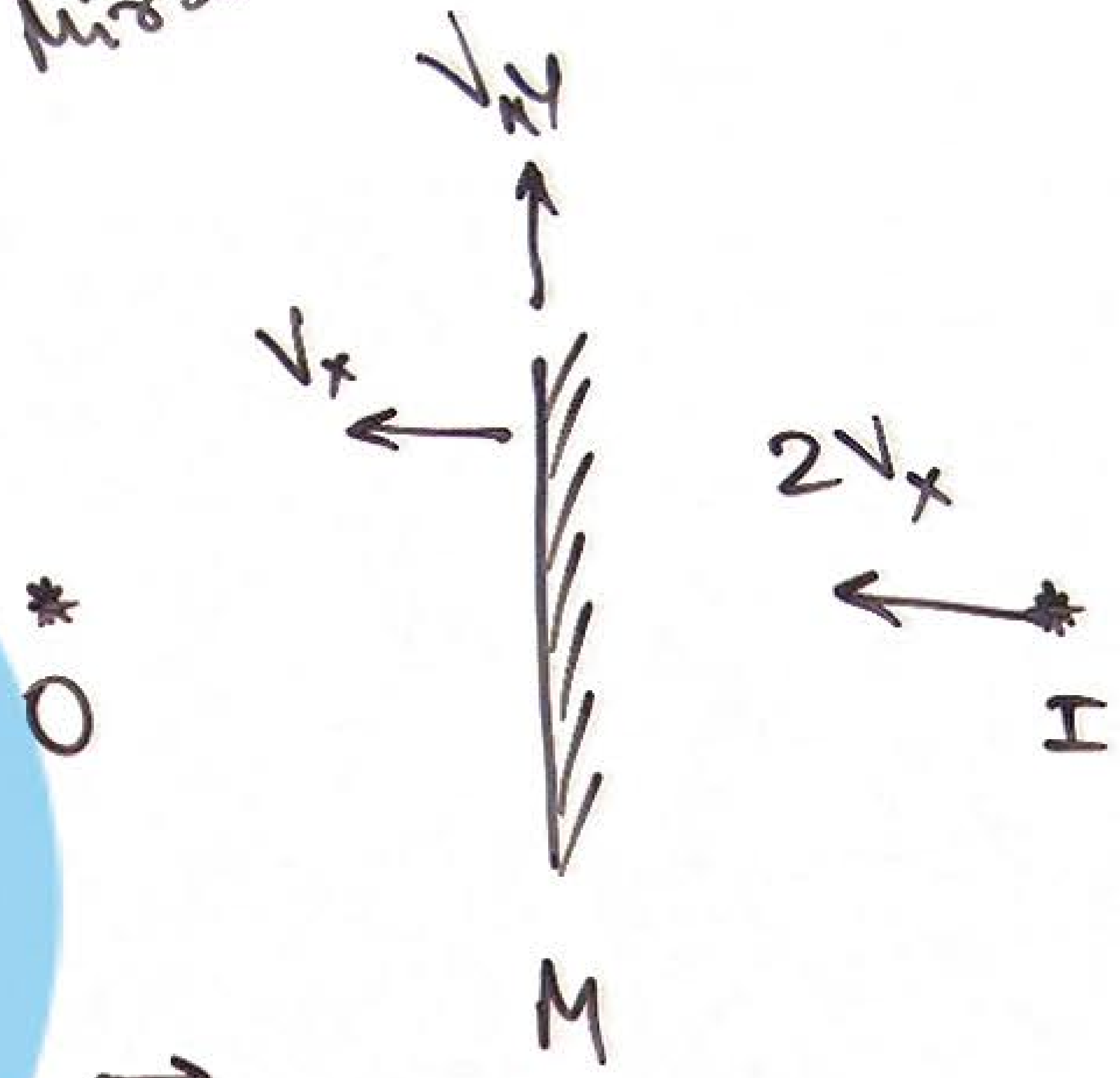
⑧ Velocity of Image by a Plain Mirror:

Case-1: When object is moving



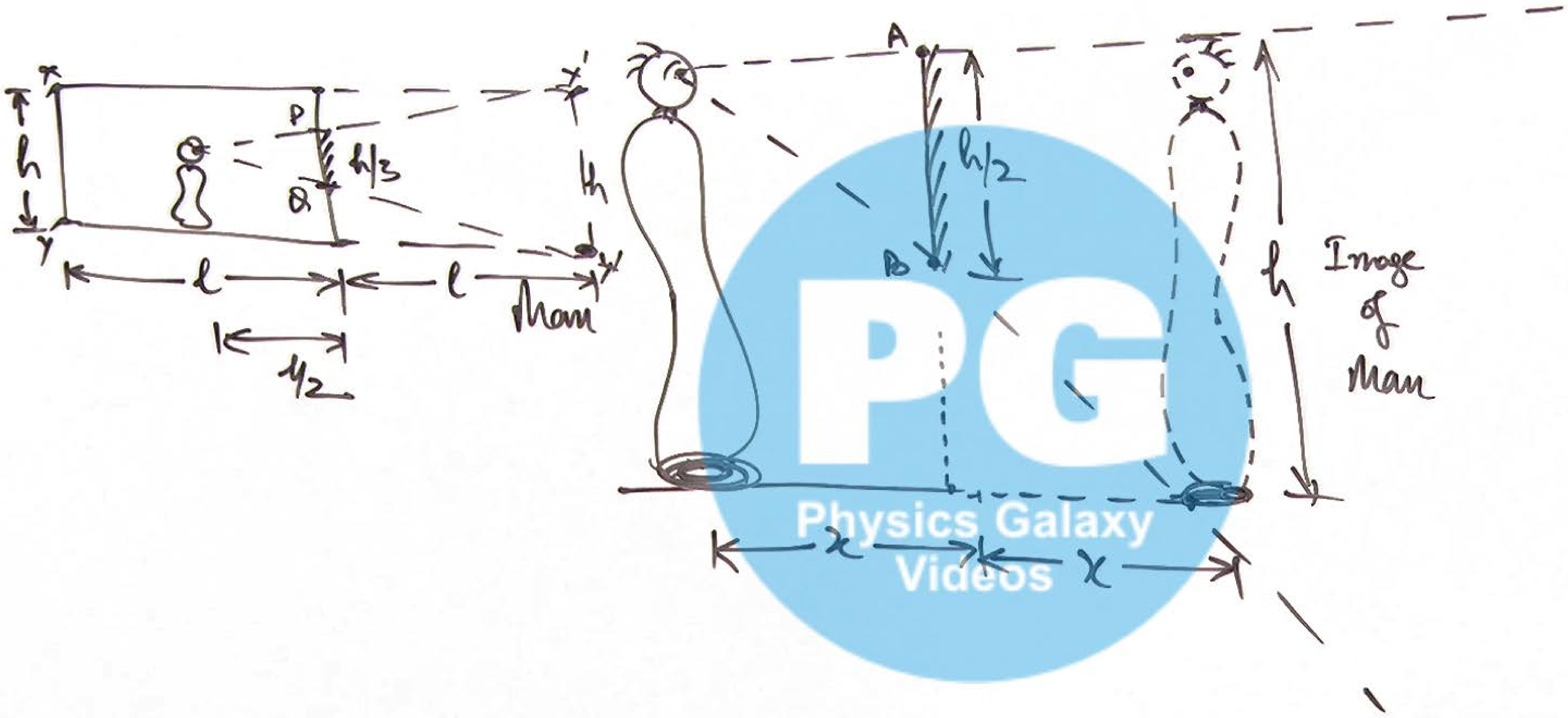
$$\left[\begin{aligned}\vec{V}_{Iy} &= \vec{V}_{oy}\\ \vec{V}_{Ix} &= -\vec{V}_{ox}\end{aligned}\right]$$

Case-2: When Mirror is moving



$$\left[\begin{aligned}\vec{V}_{Iy} &= 0\\ \vec{V}_{IX} &= 2\vec{V}_{MX}\end{aligned}\right]$$

(9) A Man requires half sized Mirror to see his full image :

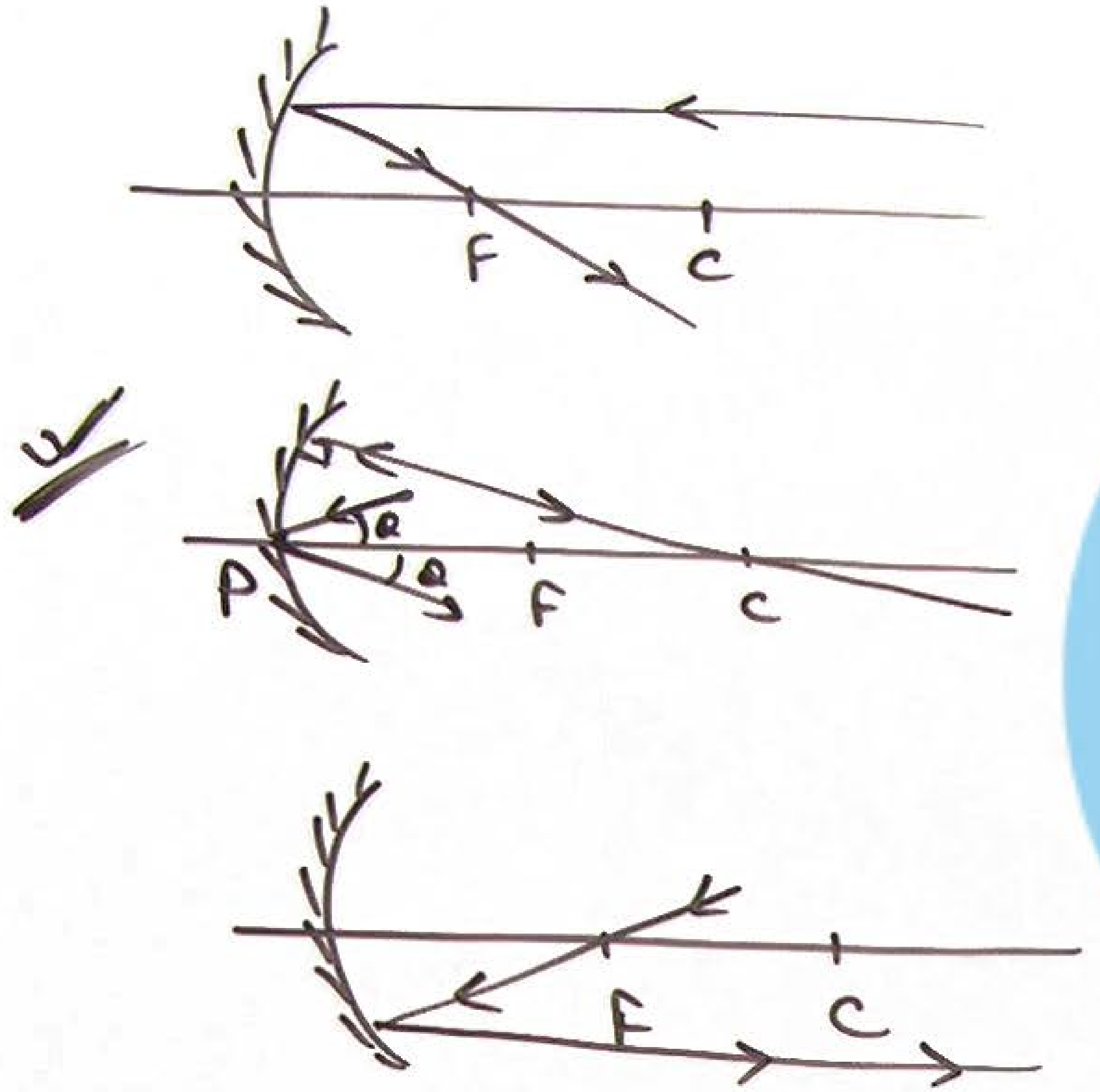


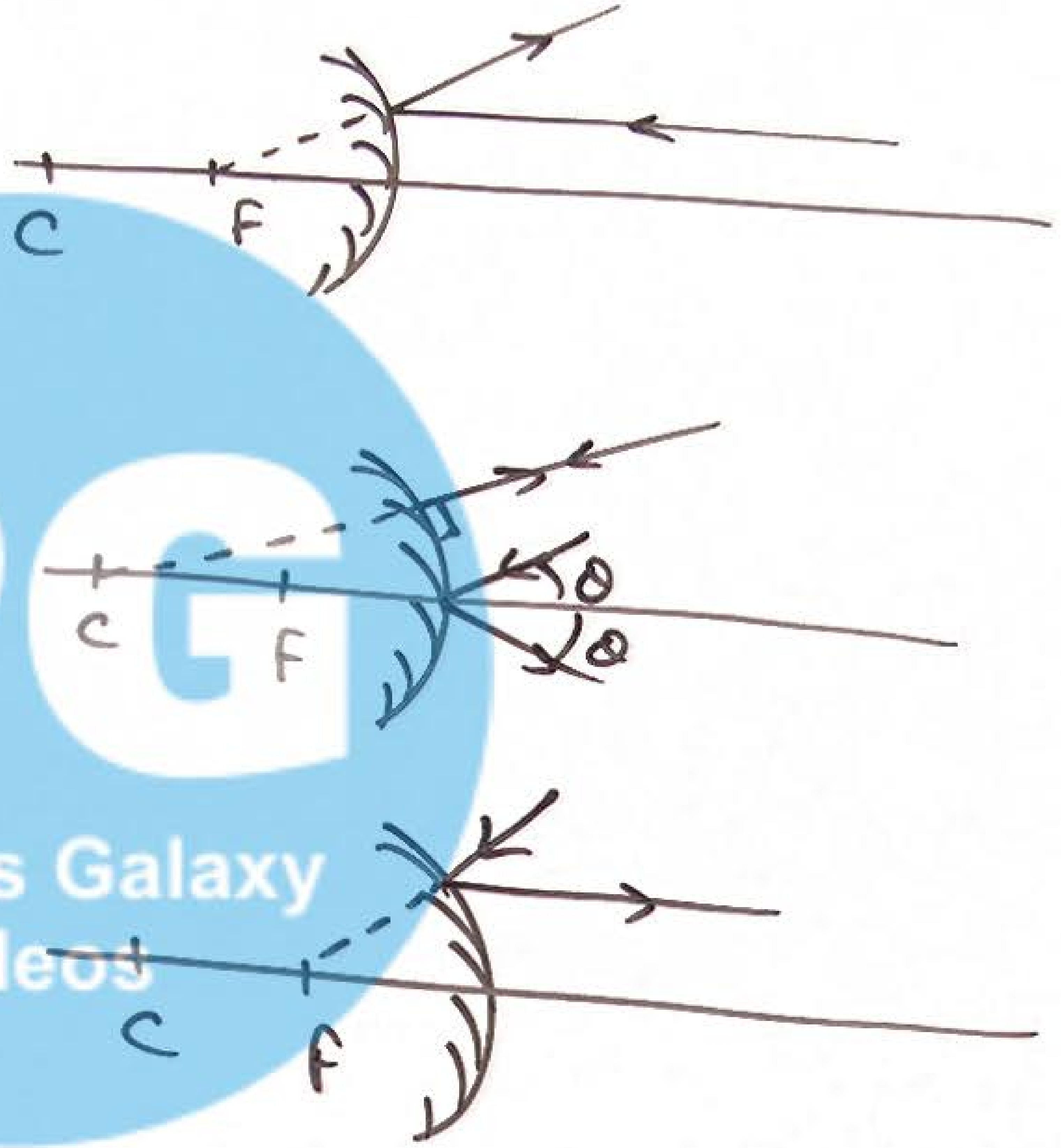
(10) Reflection by Spherical Mirrors:

for ‖ rays

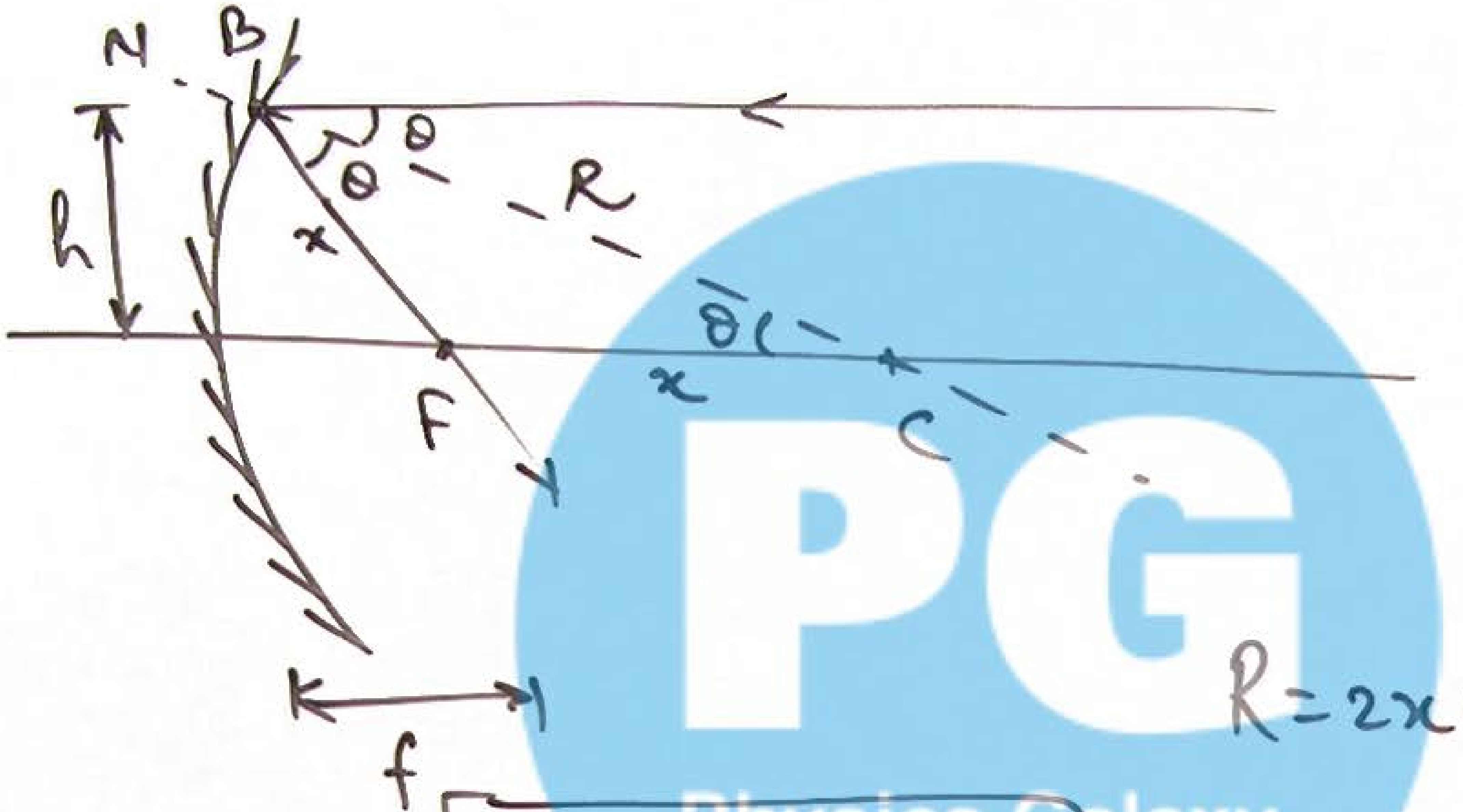
Concave Mirror (Converging M)

Convex Mirror (Diverging Mirror)





(11) Focal point for Marginal Rays:



$$\boxed{f = R - x = R - \frac{R}{2\cos\theta}}$$

$R = 2x\cos\theta$

$x = \frac{R}{2\cos\theta}$

for paraxial Rays

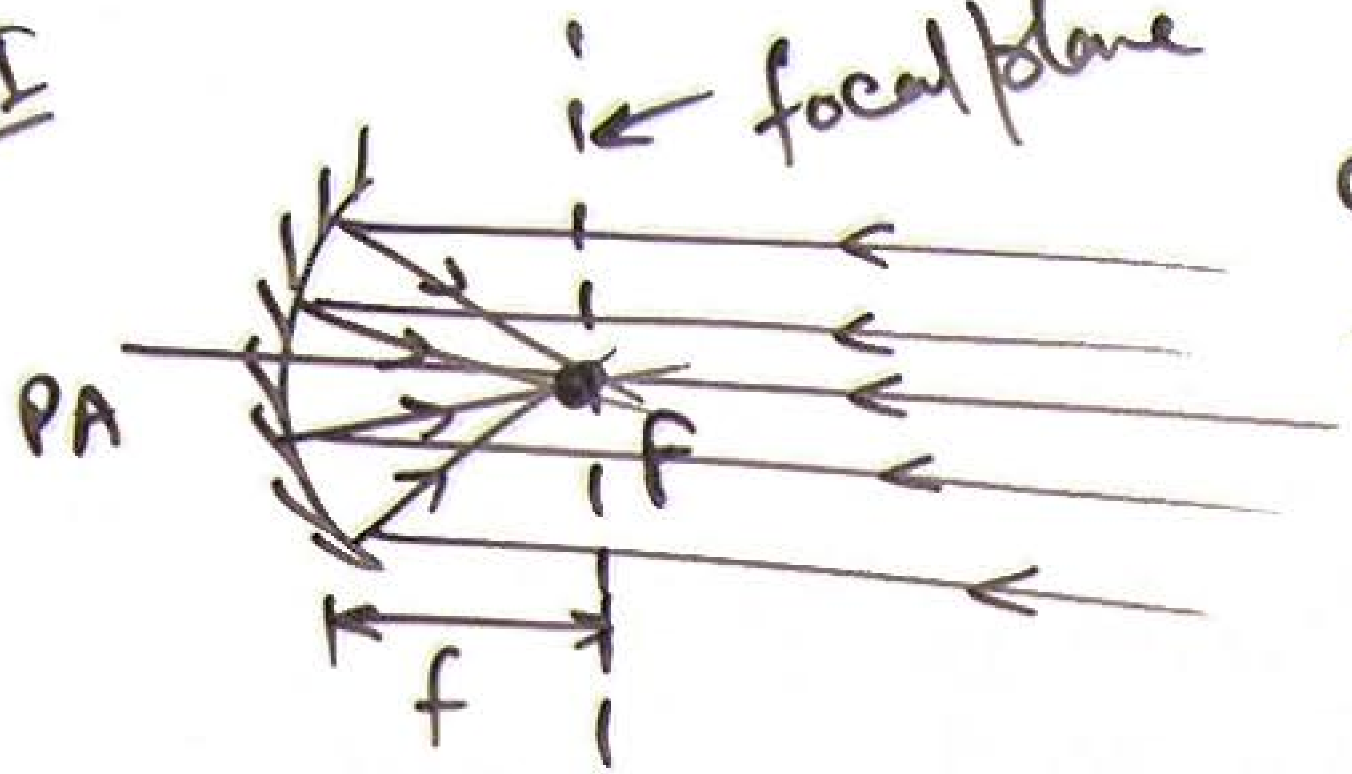
$\theta \simeq 0$

$\Rightarrow \cos\theta \simeq 1$

$\Rightarrow f = R - R/2 = \frac{R}{2}$

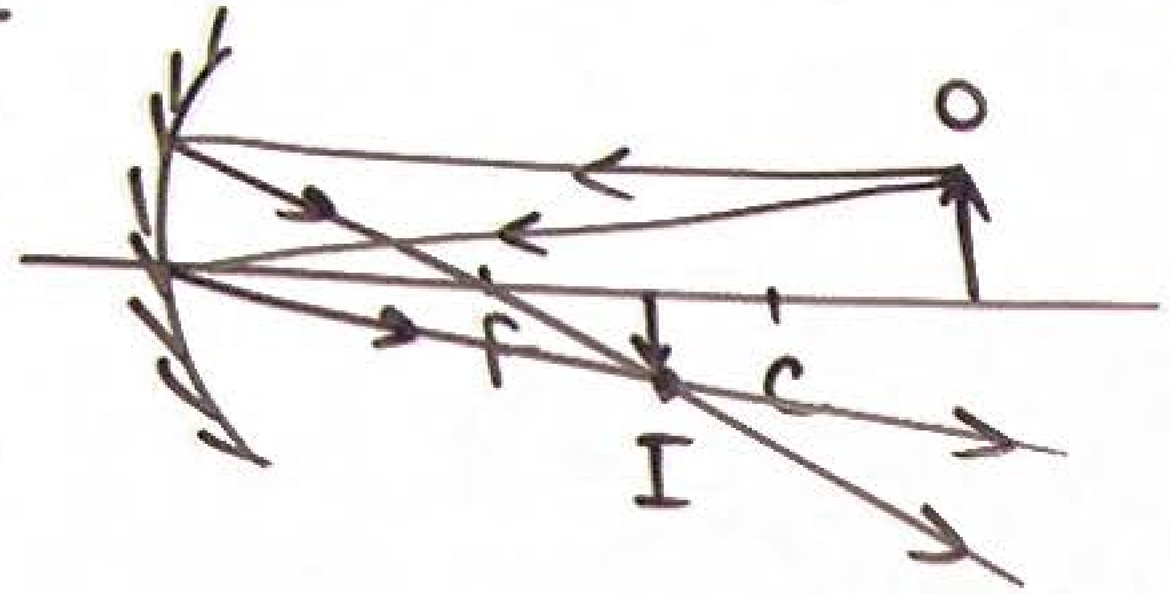
(12) Image formation by Concave Mirrors:

Case-I



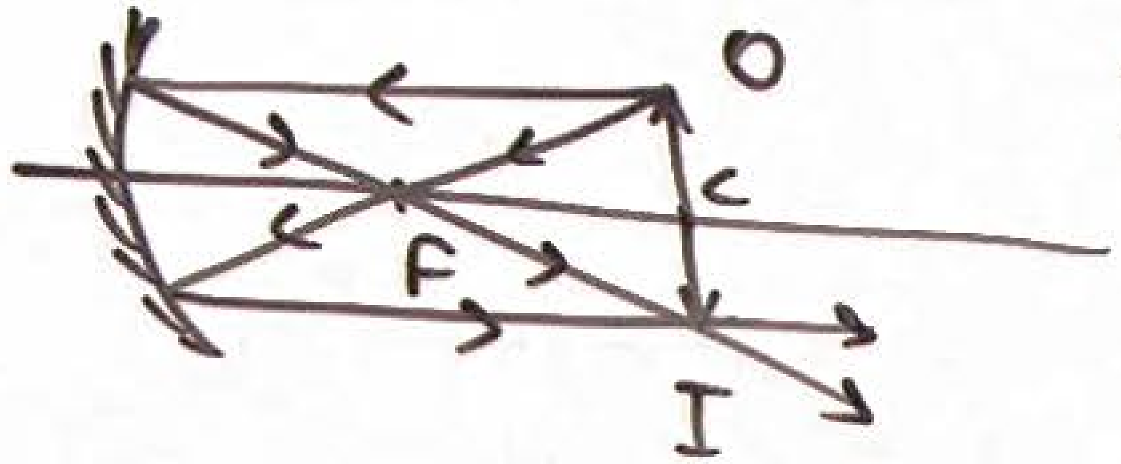
O at ∞.
I at focal plane
& pt sized.

Case-II



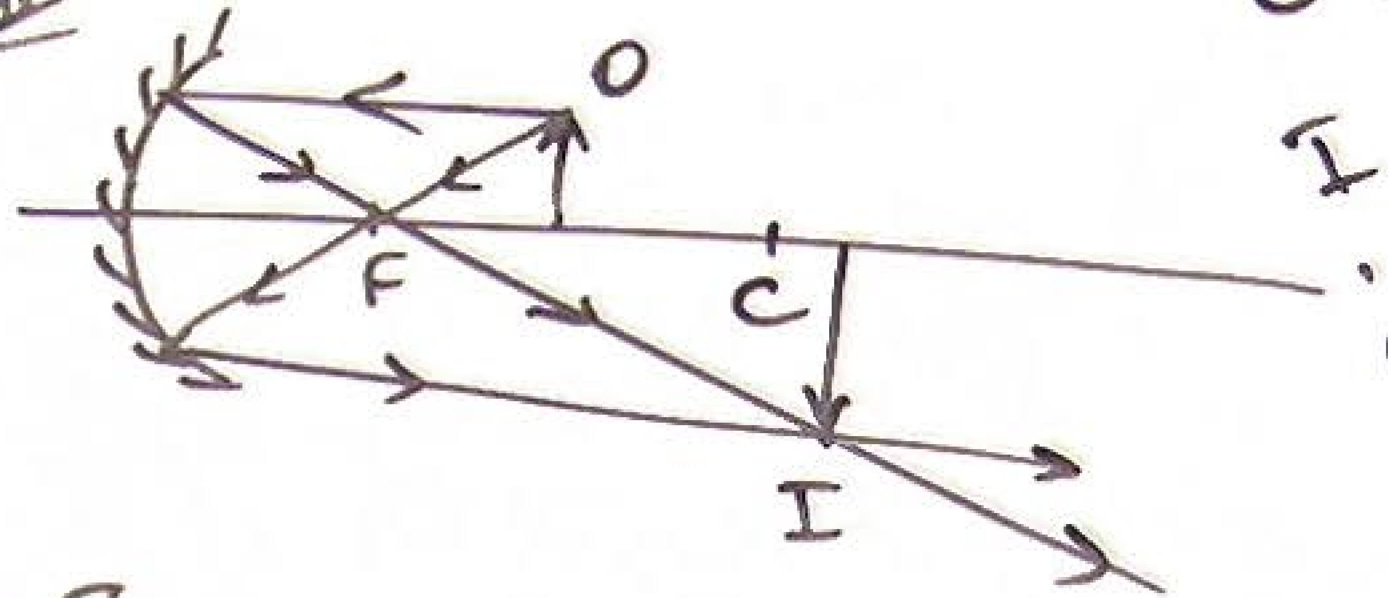
O at beyond C
I betw^n F & C
$h_i < h_o$

Case-III



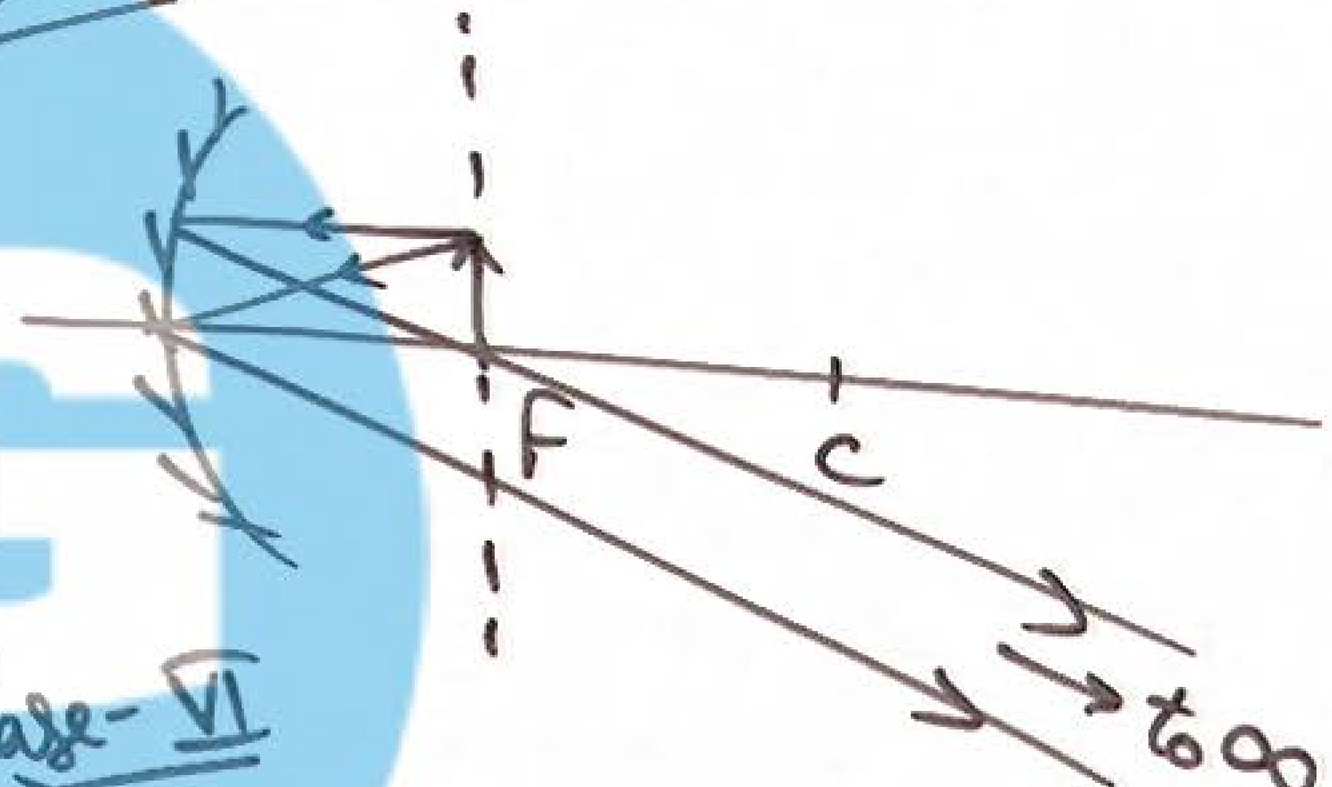
O at C
I at C
Same sized

Case-IV



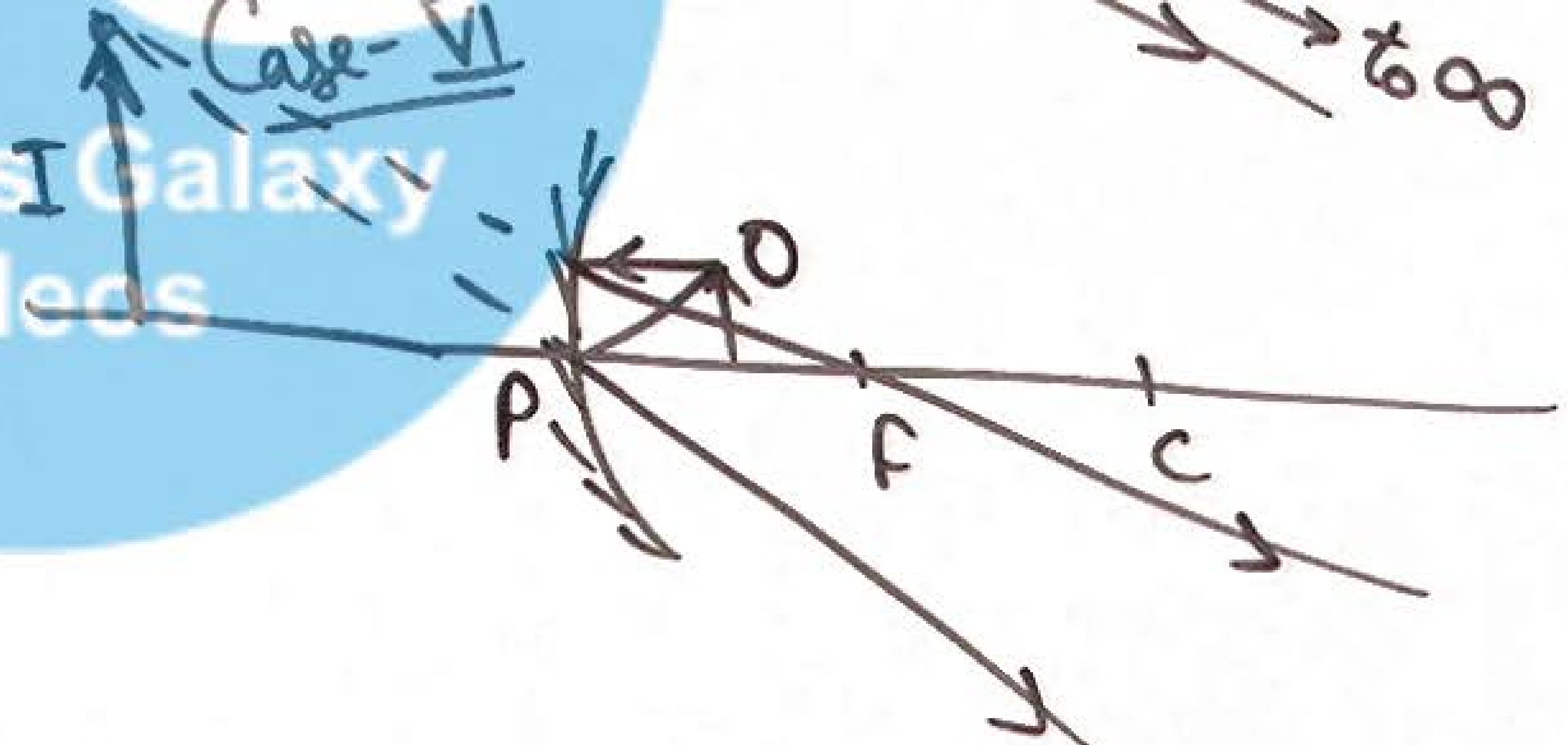
O betw^n F & C
I beyond C
'enlarged'

Case-V



O at F or in F. plane
I at ∞

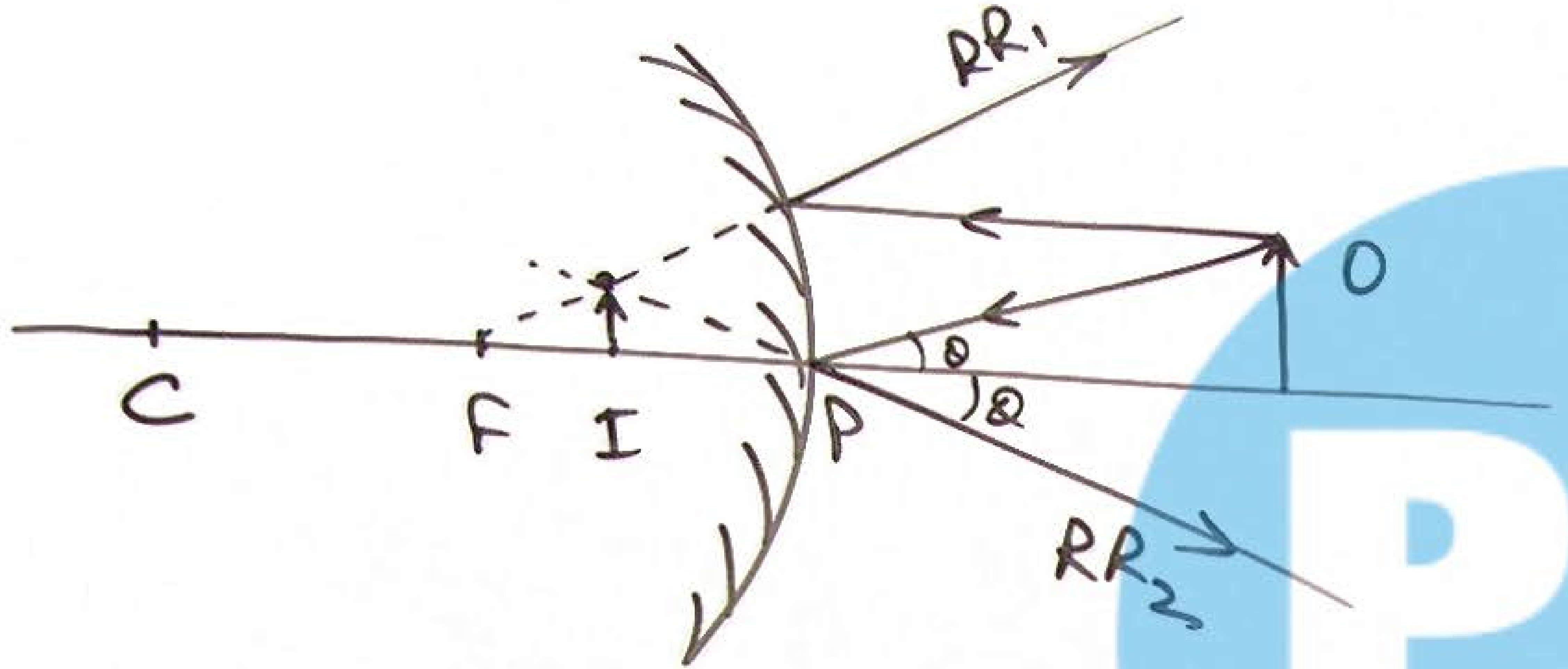
Case-VI



O betw^n F & P
I virtual
(enlarged)

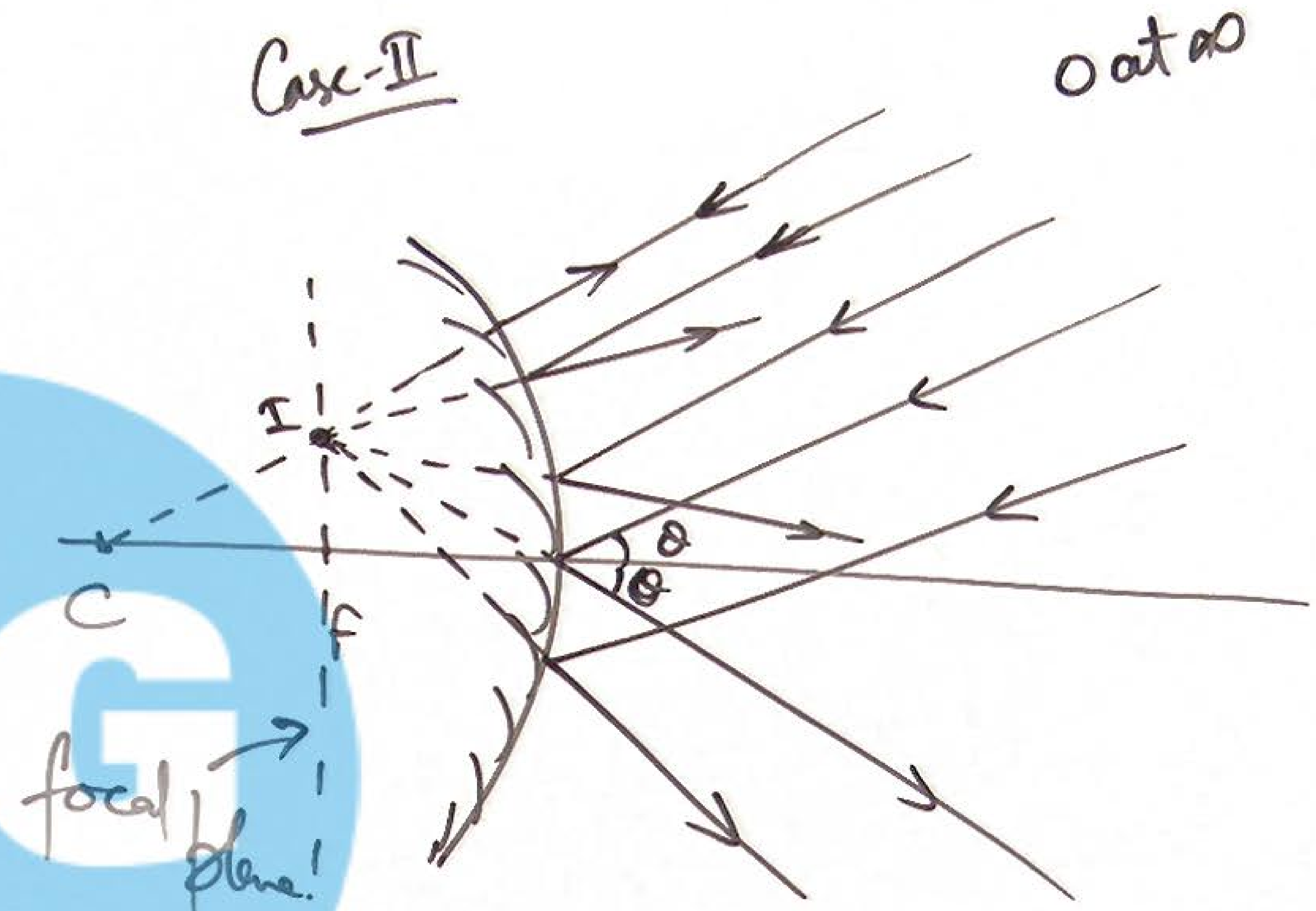
12 Image formation by Convex Mirrors:

Case-I: object is placed anywhere on PA



Virtual Image
between P & F
diminished.

Case-II

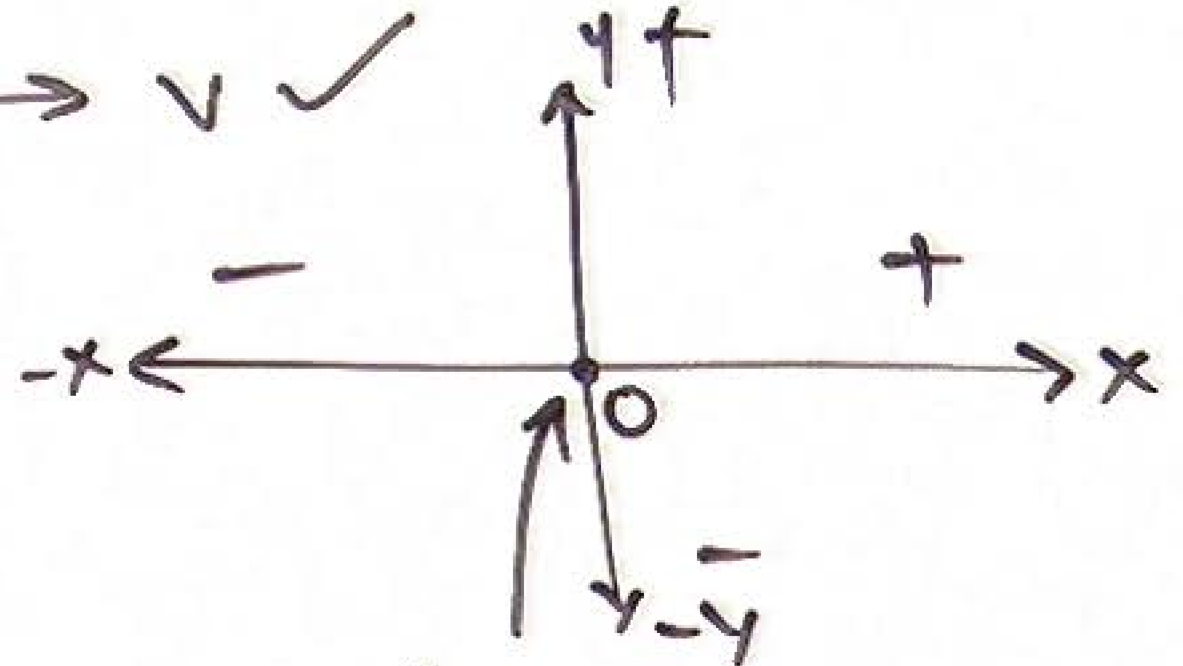


(14) Analysis of Image formation by Spherical Mirrors:

Image Parameters -

$$\boxed{\frac{1}{f} = \frac{1}{u} + \frac{1}{v}}$$

$u, f \longrightarrow v$ ✓



Pole of device.

① Location of Image
② Nature of Image
} → Mirror formula.

③ Size of Image
④ Orientation of Image
} → Magnification formula.

$$\boxed{m = \left|\frac{v}{u}\right|} \longrightarrow \text{height of image} = m \times (\text{height of object})$$

(15) Alternative forms of Mirror Formula:

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

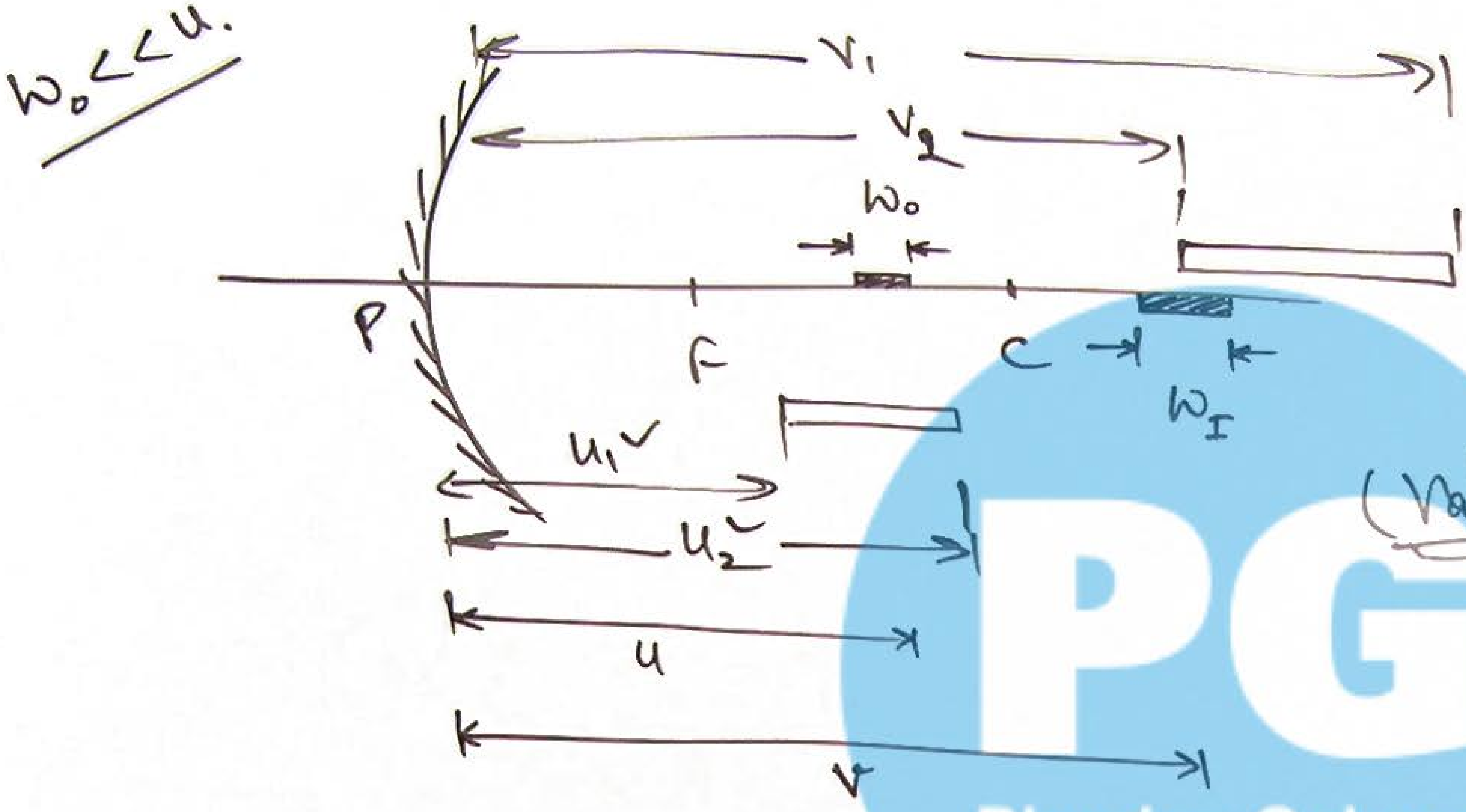
$$\rightarrow f = \frac{uv}{u+v}$$

$$\rightarrow v = \frac{uf}{u-f}$$

$$\rightarrow u = \frac{vf}{v-f}$$

$$m = -\left(\frac{v}{u}\right) = -\left(\frac{f}{u-f}\right) = -\left(\frac{v-f}{f}\right)$$

(16) Longitudinal Magnification:



longitudinal magnification

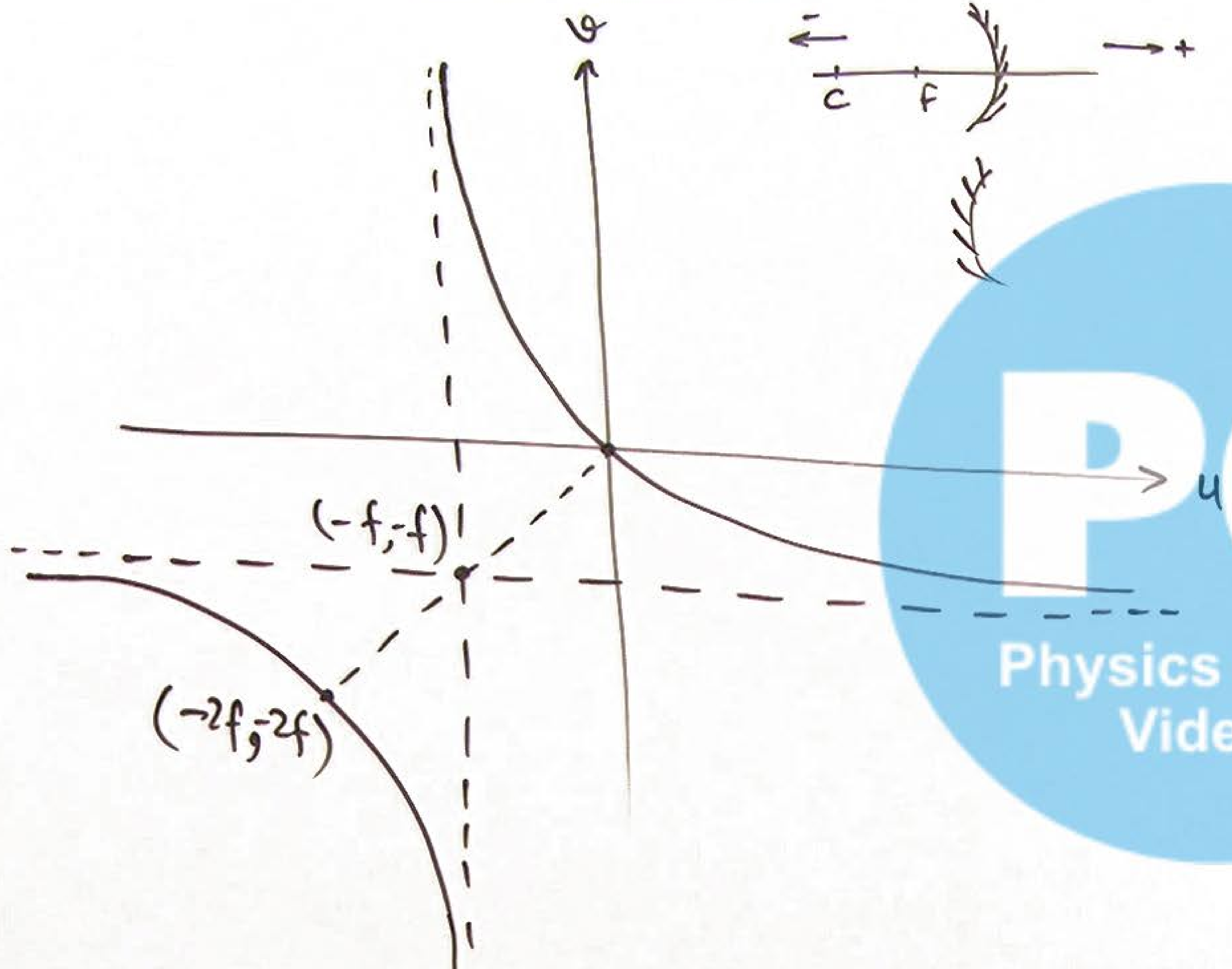
$$\rightarrow m_L = \frac{v^2}{u^2} = m^2$$

(Valid only for v. small sized objects)

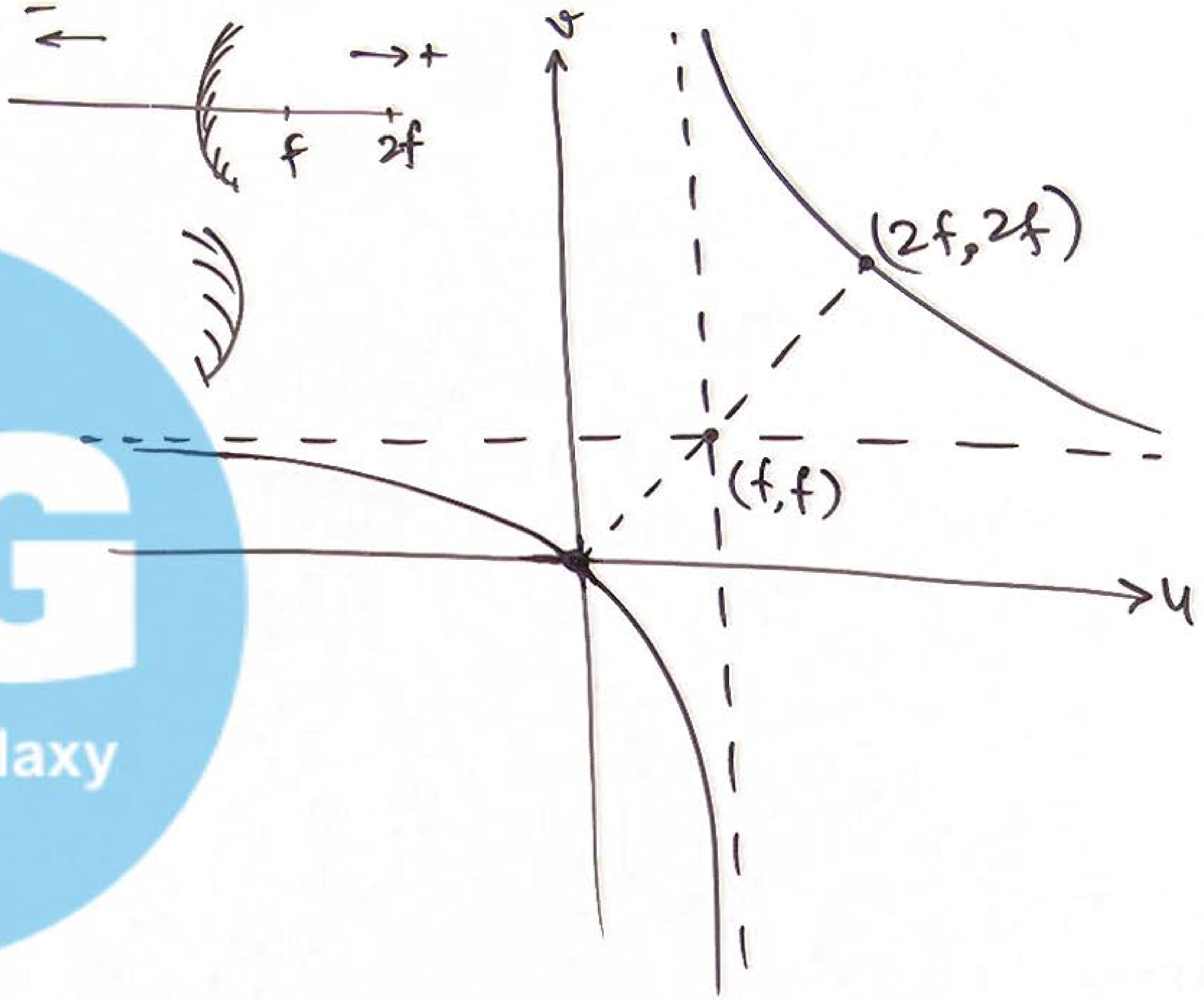
width of image $= m_L \times$ (width of object)

(17) Variation curves of Image & Object Distances:

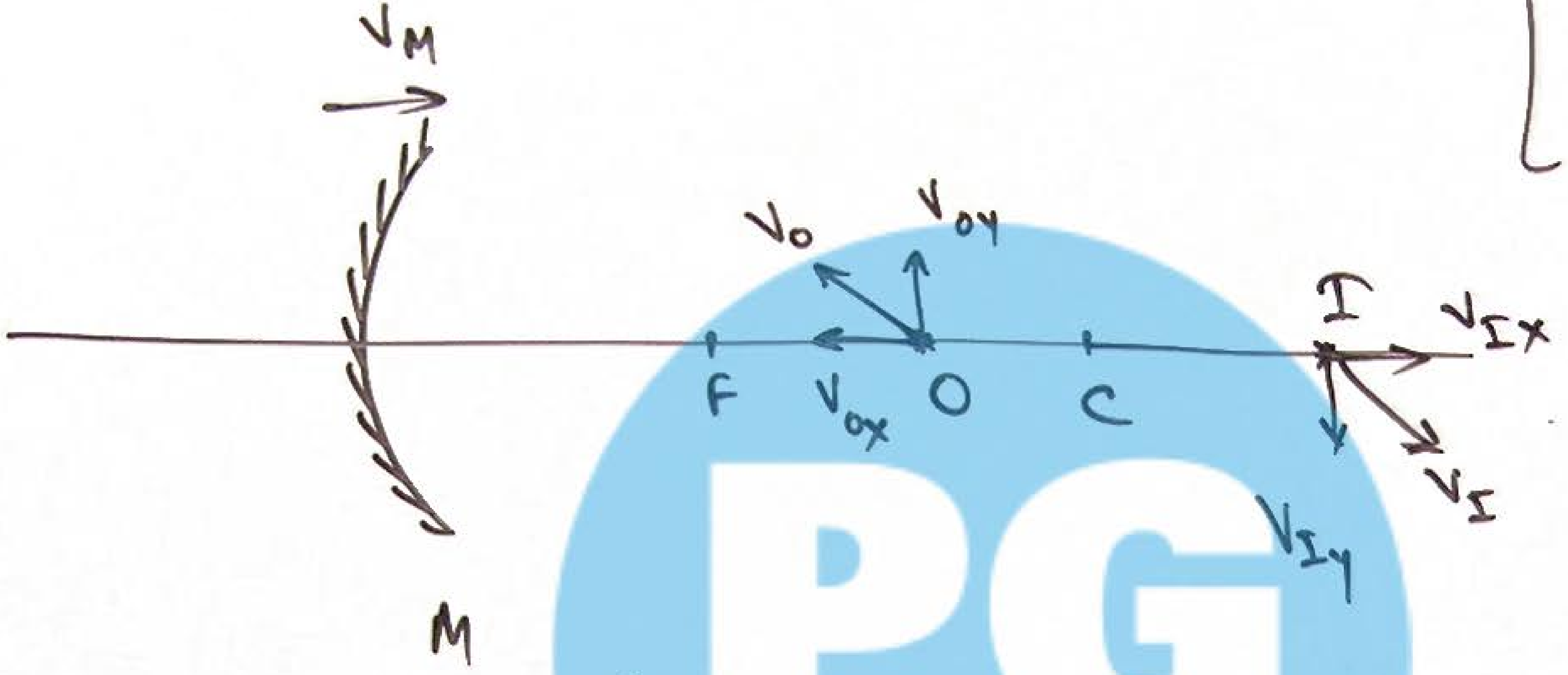
Concave Mirror



Convex Mirror



(18) Effect of motion of object & mirror on Image:



for object motion

for y dir

$$V_{Iy} = m \cdot V_{oy}$$

$$\vec{V}_{Iy} = -\frac{v}{u} \cdot \vec{V}_{oy}$$

$m = \left|\frac{v}{u}\right|$

$m = -\frac{v}{u}$

or use nature of O & I to check dir of I motion

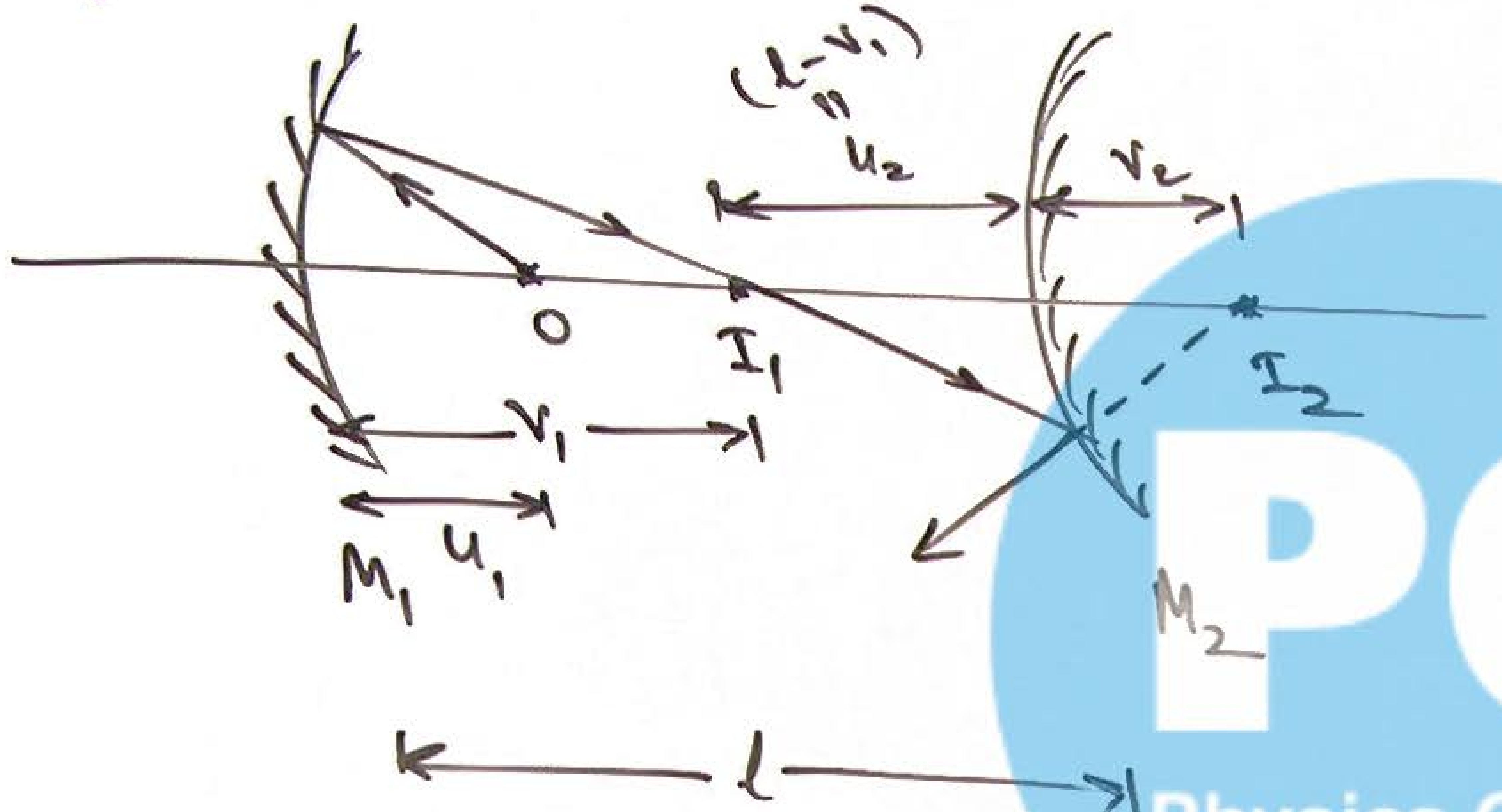
for x-dir

$$|V_{Ix}| = m^2 |V_{ox}|$$

If mirror is moving then use object & image velocity w.r.to mirror.

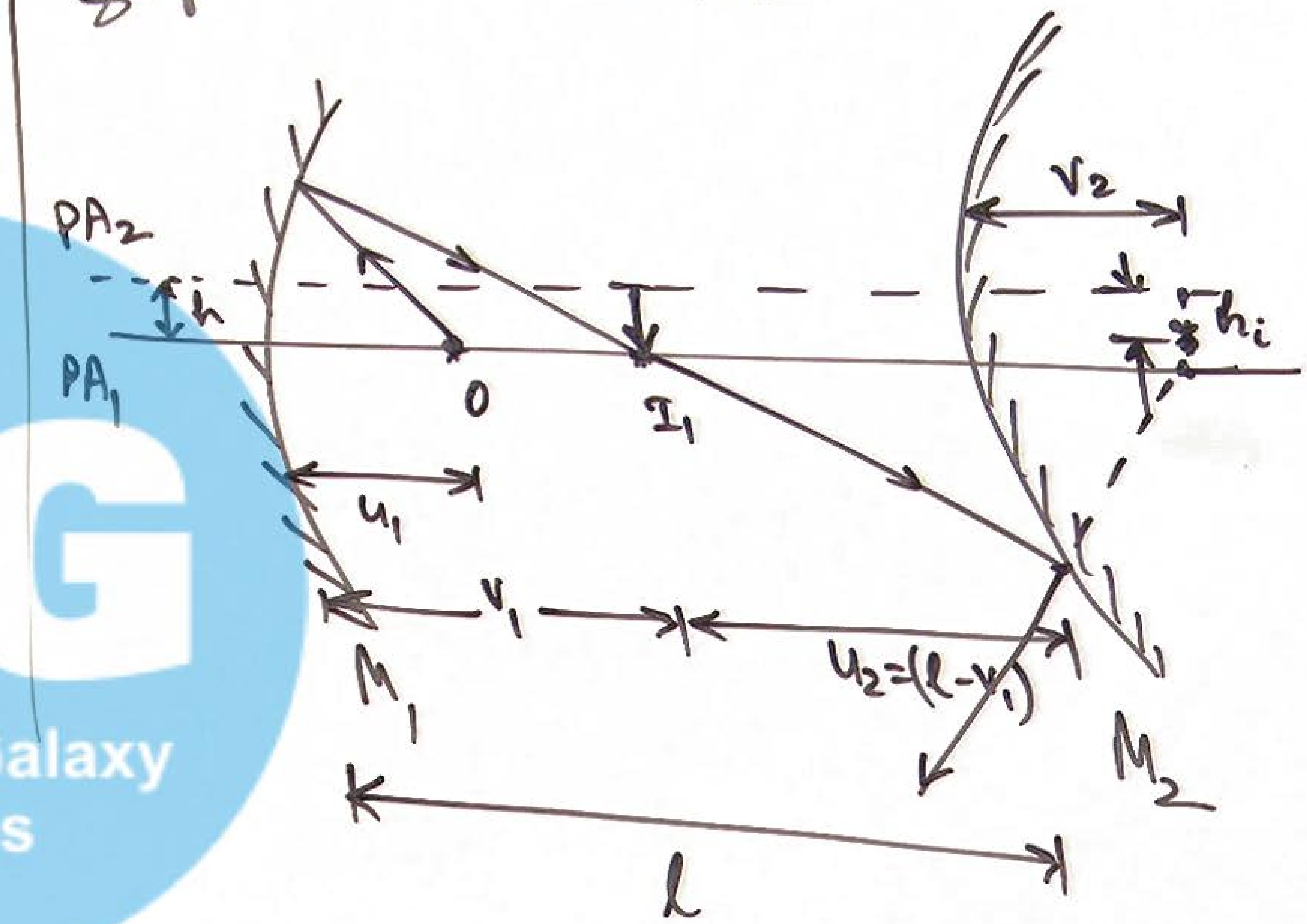
(19) Effect of Shifting PA of Mirror:

Case-I: M_1, M_2 are on same PA.

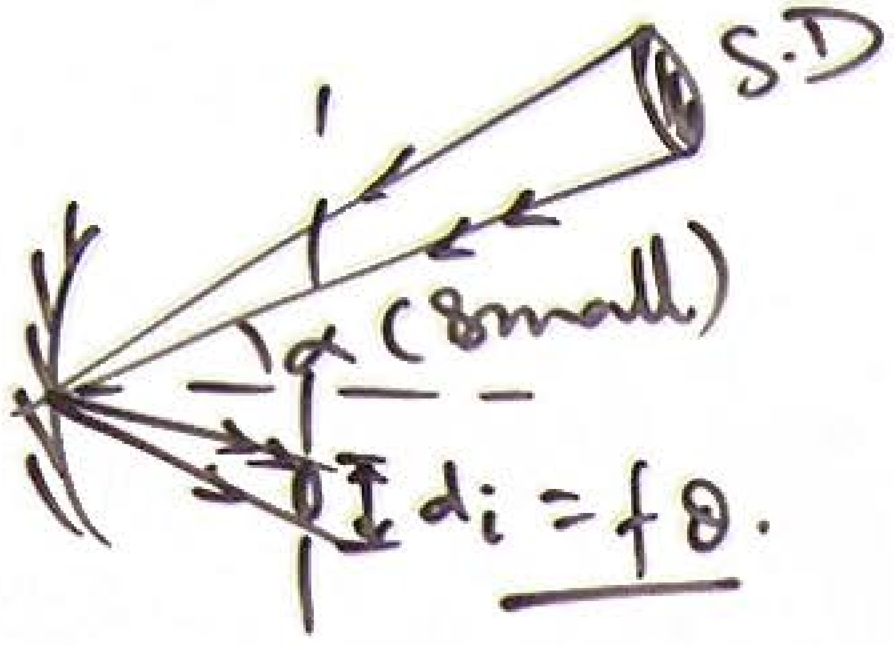


Case-II: M_2 has a shifted PA.

$$h_i = \left|\frac{v_2}{u_2}\right| \times h.$$



20 Image formation of Distant objects:



$\boxed{d_i = f\theta.}$